

A JOINT-DISTRIBUTION ROUTE TO FAIR REPRESENTATIONS WITH CONTINUOUS SENSITIVE ATTRIBUTES*

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Abstract. Fair representation learning seeks a representation Z that obscures a sensitive attribute S while preserving information about a target Y . For categorical S , this enforces parity among conditional distributions $P_{Z|S=s}$. For continuous S , existing criteria evaluate an expected conditional discrepancy over the marginal measure of S , requiring nonparametric conditional estimation.

We instead quantify independence by the discrepancy between the joint distribution $P_{Z,S}$ and the product of its marginals $P_Z \otimes P_S$. We disintegrate this joint discrepancy into a P_S -weighted conditional contrast and recover the conditional-integral IPM under decomposable witness classes. The P_S -weighting is intrinsic, and both approaches target the same independence condition without requiring conditional density estimation.

Instantiating this joint discrepancy with the Hilbert–Schmidt Independence Criterion (HSIC) yields a closed-form V -statistic estimator that requires no localization bandwidth for estimating $P_{Z|S=s}$; for learned encoders, a uniform HSIC concentration result gives root- n -type train-to-population control over controlled encoder classes, including bounded linear encoders and fixed-architecture bounded MLPs. The resulting optimization objective, FRHSIC, achieves comparable fairness–accuracy tradeoffs, transfers fairness to downstream predictors without retraining, and accelerates training by a factor of 36 per epoch on standard benchmarks.

Key words. fair representation learning, continuous sensitive attributes, Hilbert–Schmidt independence criterion, kernel methods, demographic parity

MSC codes. 68T05, 62G05, 62G20, 46E22

1. Introduction. Machine learning models increasingly make decisions that affect people in domains such as hiring, lending, and criminal justice (Calders and Verwer, 2010). A central fairness requirement is that such decisions not depend on a sensitive attribute S (e.g., race, gender, age). Fair representation learning enforces this requirement at the representation level: an encoder $h : \mathcal{X} \rightarrow \mathcal{Z}$ produces a representation $Z = h(X)$ such that downstream prediction heads $f : \mathcal{Z} \rightarrow \mathcal{Y}$ inherit fairness without imposing additional fairness constraints (Zemel et al., 2013; Louizos et al., 2015; Madras et al., 2018). At the representation level, the Demographic Parity (DP) objective $Z \perp S$ requires the distribution of Z to remain invariant across values of the sensitive attribute S .

When S takes finitely many values, the independence condition $Z \perp S$ reduces to matching the conditional distributions $P_{Z|S=s}$ across all groups. Translating this theoretical condition into an optimization objective requires a computable divergence between probability measures. Integral Probability Metrics (IPMs) provide this mechanism by defining distributional distances through differences of expectations rather than probability densities. By operating directly on empirical averages, IPMs bypass density estimation entirely. Consequently, formulations such as Maximum Mean Discrepancy (MMD) (Gretton et al., 2012) yield closed-form V -statistic estimators with root- n convergence rates under standard regularity conditions.

When S is continuous, the independence condition $Z \perp S$ requires controlling the conditional law $P_{Z|S=s}$ over a continuum of sensitive values. The population

*Submitted to the editors May 18, 2026.

Funding: This work was supported in part by NSF grant 2229876, the A. Russell Chandler III Professorship at Georgia Institute of Technology, an NIH-sponsored Georgia Clinical & Translational Science Alliance, and the Georgia Department of Transportation.

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object is well defined, but its empirical counterpart is not: finite samples contain no reliable observations at an exact value $S = s$, and repeated observations at the same sensitive value occur with probability zero. Existing criteria resolve this obstacle by replacing exact conditioning with localized conditioning over neighborhoods of s . GDP (Jiang et al., 2022) is a fairness criterion based on the local first moment $m(s) = \mathbb{E}[\hat{Y} \mid S = s]$ rather than the full conditional law; its empirical estimators approximate $m(s)$ either by hard grouping, which partitions S into histogram bins, or by soft grouping, which assigns kernel weights to samples according to their distance from the target sensitive value. EIPM (Kong et al., 2025) extends this localization idea from conditional moments to conditional distributions: for each observed S_i , it replaces $P_{Z|S=S_i}$ by a leave-one-out weighted empirical measure

$$\hat{P}_{Z|S=S_i}^{(-i),\gamma} = \sum_{j \neq i} \frac{K_\gamma(S_j, S_i)}{\sum_{k \neq i} K_\gamma(S_k, S_i)} \delta_{Z_j},$$

so that observations with S_j close to S_i serve as surrogates for the unavailable conditional sample at S_i . FREM then chooses the RKHS-unit-ball IPM, yielding an MMD-based closed-form estimator. Mutual-information-based objectives (Cho et al., 2020b) correspond formally to the KL choice in

$$(1.1) \quad \mathcal{I}_d(Z; S) := \mathbb{E}_S [d(P_{Z|S}, P_Z)],$$

but their empirical implementation requires estimating a density, density ratio, variational bound, or KDE-based surrogate rather than directly forming exact conditional empirical distributions. Across these criteria, exact conditioning on $S = s$ is replaced by a localized or density-based surrogate. This nonparametric step introduces tuning choices and, for smoothing-based GDP and EIPM/FREM estimators, leads to rates below the root- n benchmark.

We propose a joint-distribution route that avoids estimating the conditional laws $P_{Z|S=s}$. Instead of constructing local conditional distributions over neighborhoods of s , we measure fairness by comparing the joint law $P_{Z,S}$ with the product law $P_Z \otimes P_S$, whose equality is equivalent to $Z \perp S$. Theorem 3.3 disintegrates this joint discrepancy into a P_S -weighted conditional contrast, and Corollary 3.5 recovers the conditional-integral functional $\mathcal{I}_d(Z; S)$ (1.1) when the witness class decomposes across s . Thus the averaging over sensitive values is induced by the joint law itself, and the joint formulation targets the same independence condition as the conditional-integral formulation without constructing the family $\{P_{Z|S=s}\}_{s \in \mathcal{S}}$.

The HSIC instance gives a computable fairness penalty from paired samples. We instantiate the joint discrepancy with the Hilbert–Schmidt Independence Criterion (HSIC) (Gretton et al., 2005), equivalently the squared MMD between $P_{Z,S}$ and $P_Z \otimes P_S$ under a product kernel. Its empirical estimator is a closed-form $O(n^2)$ V -statistic computed from the Gram matrices of $\{Z_i\}_{i=1}^n$ and $\{S_i\}_{i=1}^n$; for a fixed encoder it converges to population HSIC at rate $O_p(n^{-1/2})$ under bounded-kernel conditions, and a uniform concentration result extends this root- n -type control to learned encoders over controlled-complexity classes. We further prove that HSIC controls kernel-smoothed conditional distribution deviations, and, under spectral regularity of the sensitive-attribute kernel, controls the conditional MMD integral up to an explicit spectral tail. The finite-sample analogue gives a projected empirical prediction-deviation bound with an explicit sensitive-kernel spectral constant. Hence the estimator replaces conditional smoothing by a single joint dependence statistic:

it requires no bins or localization bandwidth for estimating $P_{Z|S=s}$, no leave-one-out conditional empirical measures, and no covariance-operator regularization.

FRHSIC is the resulting minibatch regularizer for fair representation learning. We add the empirical HSIC penalty between the learned representation $Z = h(X)$ and the sensitive attribute S to the prediction loss. We evaluate the method on synthetic data and five real datasets: Adult, ACS Income, MEPS, Crime, and COMPAS. FRHSIC achieves comparable fairness-accuracy tradeoffs to integral-route methods, including FREM, Reg-GDP, and adversarial baselines, while training faster in our experiments. At $n = 20,000$ and batch size 256, FRHSIC trains roughly 36 times faster per epoch than FREM. Predictors trained afterward on the learned representations also retain fairness without additional downstream fairness penalties.

1.1. Summary of Contributions.

A joint formulation that avoids conditional estimation. We show that continuous-sensitive fairness can be formulated as a discrepancy between $P_{Z,S}$ and $P_Z \otimes P_S$, rather than by explicitly estimating $P_{Z|S=s}$ for each sensitive value. Theorem 3.3 connects this joint discrepancy to the P_S -weighted conditional-integral objective used by existing criteria. This result places GDP, EIPM/FREM, and mutual-information-based objectives in a common joint-vs-product perspective while preserving the target independence condition $Z \perp S$.

A root- n -type HSIC estimator with nonzero-regime control. The HSIC instance yields a closed-form $O(n^2)$ V -statistic computed from paired samples $\{(Z_i, S_i)\}_{i=1}^n$. For a fixed encoder it has the standard root- n concentration of bounded V -statistics; for learned encoders, a uniform HSIC concentration result over the feasible encoder class gives, under fixed-effective-complexity conditions that include bounded linear encoders and fixed-architecture bounded MLPs, a train-to-population error of $\tilde{O}_p(n^{-1/2})$. Beyond zero-level equivalence, we prove that HSIC controls kernel-smoothed conditional distribution deviations and, under spectral regularity of the sensitive-attribute kernel, the conditional MMD integral up to an explicit spectral tail; the finite-sample analogue is a projected empirical prediction-deviation bound with an explicit sensitive-kernel spectral constant. The estimator avoids conditional smoothing, leave-one-out conditional estimates, and covariance-operator regularization.

A minibatch training algorithm for fair representations. We implement the HSIC estimator as a minibatch regularizer, called FRHSIC. Although FRHSIC is not uniformly the best method on any single dataset, experiments on synthetic data and five real datasets support the conclusion that its fairness-accuracy tradeoff is comparable to FREM, Reg-GDP, and adversarial baselines (best on COMPAS, mid-pack elsewhere by area under the Pareto frontier), while it admits root- n -type uniform control under controlled encoder complexity and trains roughly 36 times faster per epoch (at $n = 20,000$, batch size 256); the learned representations also support downstream predictors without additional fairness penalties.

2. Preliminaries and Existing Integral Criteria.

2.1. Setup and representation-level fairness. Representation-level fairness asks for a representation that is statistically independent of the sensitive attribute while remaining predictive of the target. Let $X \in \mathcal{X}$ denote the input random vector, $S \in \mathcal{S} \subseteq \mathbb{R}$ the continuous sensitive attribute, and $Y \in \mathcal{Y}$ the target variable. A representation function $h : \mathcal{X} \rightarrow \mathcal{Z}$ yields $Z = h(X)$, and a prediction head $f : \mathcal{Z} \rightarrow \mathcal{Y}$ yields $\hat{Y} = f(Z) = f \circ h(X)$. Write P_Z for the marginal law of Z , $P_{Z|S=s}$ for the

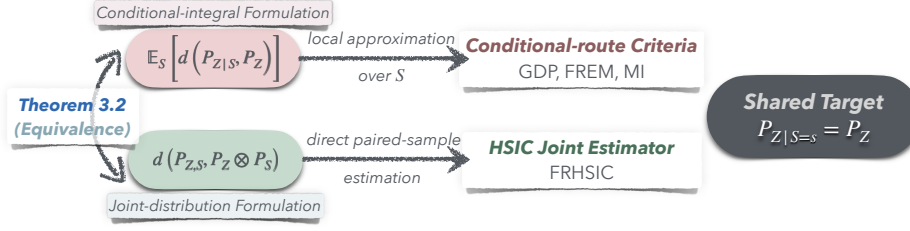


FIG. 1. *Conditional and joint routes for continuous-sensitive fair representation learning. The conditional route approximates local objects indexed by $S = s$; the joint route estimates dependence directly from paired samples.*

conditional law of Z given $S = s$, and $P_{Z,S}$ for the joint law. For a reproducing kernel Hilbert space (RKHS) $\mathcal{F}$ with kernel k , the mean embedding of a distribution P is $\mu_P = \mathbb{E}_{X \sim P}[k(X, \cdot)] \in \mathcal{F}$. For binary $S \in \{0, 1\}$, a predictor $\hat{Y}$ satisfies demographic parity (DP) if $\hat{Y} \perp S$, which for binary $\hat{Y}$ reduces to $\mathbb{E}[\hat{Y} | S = 0] = \mathbb{E}[\hat{Y} | S = 1]$. The goal of fair representation learning is a representation $Z = h(X)$ that is independent of S while remaining informative about Y . For continuous S the exact representation-level target is

$$(2.1) \quad Z \perp S, \quad \text{equivalently} \quad P_{Z|S=s} = P_Z \quad \text{for } P_S\text{-almost every } s$$

for any regular conditional distribution $\{P_{Z|S=s}\}_{s \in S}$. This independence condition is the target that every criterion in this paper enforces; empirical criteria approximate it through a discrepancy between $P_{Z|S=s}$ and P_Z , introduced next.

2.2. Discrepancies between probability laws. Distances between distributions in this paper are integral probability metrics, and the maximum mean discrepancy is the kernel instance that vanishes exactly when two laws coincide. For a class $\mathcal{F}$ of bounded measurable functions, the integral probability metric (IPM) between distributions P and Q is $\text{IPM}_{\mathcal{F}}(P, Q) := \sup_{f \in \mathcal{F}} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]|$. Taking $\mathcal{F}$ to be the unit ball of an RKHS gives the maximum mean discrepancy,

$$\text{MMD}(P, Q) = \sup_{\|f\|_{\mathcal{F}} \leq 1} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]| = \|\mu_P - \mu_Q\|_{\mathcal{F}}.$$

When k is characteristic, $\text{MMD}(P, Q) = 0$ if and only if $P = Q$ (Gretton et al., 2012). Section 3 applies the IPM with $P = P_{Z,S}$ and $Q = P_Z \otimes P_S$.

2.3. Conditional-integral criteria for continuous sensitive attributes. Existing continuous- S fairness criteria share one form, the S -average of a per-value discrepancy between the conditional and marginal laws of Z , and differ only in that discrepancy. For a discrepancy d between distributions, the integral functional is

$$\mathcal{I}_d(Z; S) := \mathbb{E}_S[d(P_{Z|S}, P_Z)],$$

which is well-defined for Polish $\mathcal{Z}, \mathcal{S}$ once a regular conditional probability is selected; existence follows from the disintegration theorem (Kallenberg, 2002, Theorem 5.4). Three instances recur. Generalized demographic parity (GDP) (Jiang et al., 2022) takes d to be a first-moment difference evaluated through a measurable head $f : \mathcal{Z} \rightarrow \mathbb{R}$,

$$(2.2) \quad \Delta_{\text{GDP}}(f) = \mathbb{E}_S |\mathbb{E}_Z[f(Z) | S] - \mathbb{E}_Z[f(Z)]|, \quad Z = h(X),$$

with the conditional expectation estimated by Nadaraya–Watson kernel smoothing on S (Nadaraya, 1964; Watson, 1964); GDP reduces to DP when S is binary. The expectation of IPM (EIPM) (Kong et al., 2025) takes d to be an IPM,

$$\text{EIPM}_{\mathcal{V}}(Z; S) := \mathbb{E}_S[\text{IPM}_{\mathcal{V}}(P_{Z|S}, P_Z)], \quad \text{IPM}_{\mathcal{V}}(P, Q) = \sup_{f \in \mathcal{V}} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]|,$$

and controlling EIPM controls GDP (Kong et al., 2025). Mutual information is the Kullback–Leibler instance,

$$I(Z; S) = \mathbb{E}_S[\text{KL}(P_{Z|S} \| P_Z)],$$

and characterizes independence, $I(Z; S) = 0 \iff Z \perp S$ (Cover and Thomas, 2006, Theorem 2.6.3). GDP, EIPM, and mutual information are conditional-integral criteria with different local discrepancies. Section 3 shows how a joint-vs-product discrepancy is related to this conditional-integral route through disintegration, and how the conditional-integral IPM is recovered under decomposable witness classes.

2.4. Joint dependence measures. An alternative measures dependence directly between the joint law and the product of marginals, and the Hilbert–Schmidt independence criterion is the kernel instance that vanishes exactly under independence. This is the object the method of this paper is built on.

DEFINITION 2.1 (Mean embeddings and HSIC). *Let $\mathcal{F}_{\mathcal{Z}}$ and $\mathcal{F}_{\mathcal{S}}$ be RKHSs on $\mathcal{Z}$ and $\mathcal{S}$ with bounded characteristic kernels $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$, and let $\mathcal{F}_{\mathcal{Z} \otimes \mathcal{S}}$ be the tensor-product RKHS on $\mathcal{Z} \times \mathcal{S}$ with product kernel $k_{\mathcal{Z}} \otimes k_{\mathcal{S}}$. The marginal and joint mean embeddings are*

$$\mu_{\mathcal{Z}} := \mathbb{E}_{\mathcal{Z}}[k_{\mathcal{Z}}(Z, \cdot)] \in \mathcal{F}_{\mathcal{Z}}, \quad \mu_{\mathcal{S}} := \mathbb{E}_{\mathcal{S}}[k_{\mathcal{S}}(S, \cdot)] \in \mathcal{F}_{\mathcal{S}},$$

$$\mu_{\mathcal{Z}, \mathcal{S}} := \mathbb{E}_{(\mathcal{Z}, \mathcal{S})}[k_{\mathcal{Z}}(Z, \cdot) \otimes k_{\mathcal{S}}(S, \cdot)] \in \mathcal{F}_{\mathcal{Z} \otimes \mathcal{S}},$$

and $\mu_{\mathcal{Z}} \otimes \mu_{\mathcal{S}} \in \mathcal{F}_{\mathcal{Z} \otimes \mathcal{S}}$ is the embedding of the product law $P_{\mathcal{Z}} \otimes P_{\mathcal{S}}$. The conditional mean embedding of Z given $S = s$ is $\mu_{\mathcal{Z}|S=s} := \mathbb{E}[k_{\mathcal{Z}}(Z, \cdot) \mid S = s] \in \mathcal{F}_{\mathcal{Z}}$; we write $\mu_{\mathcal{Z}|S}$ for the $\mathcal{F}_{\mathcal{Z}}$ -valued random element $s \mapsto \mu_{\mathcal{Z}|S=s}$ evaluated at S . The Hilbert–Schmidt independence criterion (HSIC) is the squared maximum mean discrepancy between the joint law and the product of marginals,

$$\text{HSIC}(\mathcal{Z}, \mathcal{S}) := \|\mu_{\mathcal{Z}, \mathcal{S}} - \mu_{\mathcal{Z}} \otimes \mu_{\mathcal{S}}\|_{\mathcal{F}_{\mathcal{Z} \otimes \mathcal{S}}}^2.$$

Under characteristic product kernels, $\text{HSIC}(\mathcal{Z}, \mathcal{S}) = 0$ if and only if $Z \perp S$ (Gretton et al., 2005). Section 3 relates this joint discrepancy to conditional-integral criteria through a disintegration identity. The equality with the conditional-integral IPM requires the decomposability condition stated in Corollary 3.5.

3. Joint-Distribution Framework.

3.1. Zero-level relation to existing criteria. At zero, the characteristic-kernel property shows that HSIC recovers exactly the representation-level fairness target shared by the existing criteria. The nonzero regime is developed in §3.2–§3.4.

PROPOSITION 3.1 (Zero-level characterization of representation fairness). *Let $\mathcal{Z}$ and $\mathcal{S}$ be Polish spaces, let $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$ be bounded characteristic kernels, and let (Z, S) be a random pair on $\mathcal{Z} \times \mathcal{S}$. Then the following are equivalent:*

1. $\text{HSIC}(\mathcal{Z}, \mathcal{S}) = 0$;

2. $Z \perp S$, equivalently $P_{Z,S} = P_Z \otimes P_S$;

3. for some, equivalently every, regular conditional distribution $\{P_{Z|S=s}\}_{s \in \mathcal{S}}$,

$$P_{Z|S=s} = P_Z \quad \text{for } P_S\text{-a.e. } s;$$

4. for every bounded measurable $g : \mathcal{Z} \rightarrow \mathbb{R}$,

$$\mathbb{E}[g(Z) \mid S = s] = \mathbb{E}[g(Z)] \quad \text{for } P_S\text{-a.e. } s.$$

Proof. Conditions (1)–(4) are proved in Appendix SM1.1; the equivalence with $\text{EIPM}_{\mathcal{V}}(Z; S) = 0$ for the unit ball $\mathcal{V} \subset \mathcal{F}_{\mathcal{Z}}$ in Appendix SM1.2; and the downstream-head form in Appendix SM1.3.

Remark 3.2 (Zero-level equivalence only). Proposition 3.1 shows that HSIC, EIPM with a characteristic RKHS, mutual information, and GDP-style downstream parity agree at the zero fairness target. This is a zero-level statement. Away from zero, these criteria are different functionals: EIPM averages conditional MMD magnitudes, GDP measures head-specific conditional moments, and HSIC measures a covariance-weighted joint-vs-product discrepancy.

3.2. Conditional-integral and joint formulations. Conditional-integral criteria compare $P_{Z|S=s}$ with P_Z and average the comparison over S , whereas the joint route compares the joint law $P_{Z,S}$ with the product of marginals $P_Z \otimes P_S$. When S takes finitely many values the two formulations coincide as aggregated two-sample MMDs, recorded as Lemma SM2.2 in Appendix SM2.1 of the supplement. The case of practical interest is continuous S , where the conditional family $\{P_{Z|S=s}\}_s$ is uncountable and the two formulations differ in how they are estimated. The next theorem disintegrates the joint-vs-product IPM into a P_S -weighted conditional contrast.

THEOREM 3.3 (Disintegration of the joint-vs-product IPM). *Let $\mathcal{Z}$ and $\mathcal{S}$ be standard Borel spaces, and let $\{P_{Z|S=s}\}_{s \in \mathcal{S}}$ be a regular conditional distribution of Z given S . For any class $\mathcal{F}$ of bounded measurable functions on $\mathcal{Z} \times \mathcal{S}$,*

$$\text{IPM}_{\mathcal{F}}(P_{Z,S}, P_Z \otimes P_S) = \sup_{f \in \mathcal{F}} \left| \int_{\mathcal{S}} \int_{\mathcal{Z}} f(z, s) d(P_{Z|S=s} - P_Z)(z) dP_S(ds) \right|.$$

Proof. See Appendix SM1.4 of the supplement. $\square$

The theorem shows that the P_S weighting arises from disintegrating the joint law, rather than from an external weighting choice. It is an identity between two ways of writing the joint-vs-product IPM, not an automatic equality with the conditional-integral functional $\mathcal{I}_d$. The conditional-integral IPM places a local supremum *inside* the S -integral, whereas the joint IPM has a single supremum *outside* it; the two agree only for witness classes that decompose across s .

DEFINITION 3.4 (Decomposable witness class). *Let $\mathcal{V}$ be a symmetric class of bounded measurable functions on $\mathcal{Z}$, meaning $g \in \mathcal{V}$ implies $-g \in \mathcal{V}$. A class $\mathcal{F}$ of bounded measurable functions on $\mathcal{Z} \times \mathcal{S}$ is decomposable over $\mathcal{V}$ if:*

1. for every measurable selector $s \mapsto g_s \in \mathcal{V}$, the function $(z, s) \mapsto g_s(z)$ belongs to $\mathcal{F}$;
2. for every $f \in \mathcal{F}$, the slice $f(\cdot, s)$ belongs to $\mathcal{V}$ for P_S -almost every s .

COROLLARY 3.5 (Recovery of the conditional-integral IPM under decomposability). *Adopt the assumptions of Theorem 3.3: $\mathcal{Z}$ and $\mathcal{S}$ are standard Borel spaces and*

250 $\{P_{Z|S=s}\}_{s \in \mathcal{S}}$ is a regular conditional distribution of Z given S . Let $\mathcal{V}$ be a symmetric
 251 uniformly bounded class of measurable functions on $\mathcal{Z}$, and define

$$252 \quad d_{\mathcal{V}}(P, Q) := \text{IPM}_{\mathcal{V}}(P, Q).$$

253 Let $\mathcal{F}$ be decomposable over $\mathcal{V}$ in the sense of Definition 3.4. Assume that for every
 254 $\varepsilon > 0$ there exists a measurable selector $s \mapsto g_s^\varepsilon \in \mathcal{V}$ such that

$$255 \quad \int g_s^\varepsilon(z) d(P_{Z|S=s} - P_Z)(z) \geq d_{\mathcal{V}}(P_{Z|S=s}, P_Z) - \varepsilon \quad \text{for } P_S\text{-a.e. } s.$$

256 Then

$$257 \quad \text{IPM}_{\mathcal{F}}(P_{Z,S}, P_Z \otimes P_S) = \mathbb{E}_S[d_{\mathcal{V}}(P_{Z|S=s}, P_Z)] = \mathcal{I}_{d_{\mathcal{V}}}(Z; S).$$

258 *Proof.* See Appendix SM1.5 of the supplement. $\square$

259 GDP, EIPM, and mutual information are conditional-integral criteria with differ-
 260 ent local discrepancies d : a first-moment difference, an IPM, and the KL divergence,
 261 respectively. HSIC is a joint-distribution discrepancy, and Corollary 3.5 recovers the
 262 conditional-integral IPM exactly when the witness class decomposes across s .

263 **3.3. The HSIC joint discrepancy.** HSIC is the kernel joint discrepancy ob-
 264 tained by comparing $P_{Z,S}$ with $P_Z \otimes P_S$. Instantiating the joint IPM of Theorem 3.3
 265 with the unit ball of the tensor-product RKHS $\mathcal{F}_{\mathcal{Z} \otimes \mathcal{S}}$ of Definition 2.1 gives the max-
 266 imum mean discrepancy $\text{MMD}_{k_{\mathcal{Z}} \otimes k_{\mathcal{S}}}(P_{Z,S}, P_Z \otimes P_S)$, and HSIC is its square,

$$267 \quad \text{HSIC}(Z, S) = \text{MMD}_{k_{\mathcal{Z}} \otimes k_{\mathcal{S}}}^2(P_{Z,S}, P_Z \otimes P_S).$$

268 We assume throughout that the product kernel $k_{\mathcal{Z}} \otimes k_{\mathcal{S}}$ is characteristic to the rel-
 269 evant joint and product measures on $\mathcal{Z} \times \mathcal{S}$; this holds under standard conditions
 270 for the Gaussian kernels used in our experiments. The following theorem records the
 271 conditional-deviation decomposition of HSIC and a one-way bound by the squared
 272 expected MMD.

273 **THEOREM 3.6 (HSIC and Expected MMD).** Suppose $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$ are bounded,
 274 i.e., $k_{\mathcal{Z}}(z, z) \leq \kappa_{\mathcal{Z}}$ and $k_{\mathcal{S}}(s, s) \leq \kappa_{\mathcal{S}}$ for all $z \in \mathcal{Z}$, $s \in \mathcal{S}$. Then HSIC admits the
 275 decomposition

$$276 \quad (3.1) \quad \text{HSIC}(Z, S) = \mathbb{E}_{S, S'} \left[\langle \mu_{Z|S} - \mu_Z, \mu_{Z|S'} - \mu_Z \rangle_{\mathcal{F}_{\mathcal{Z}}} \cdot k_{\mathcal{S}}(S, S') \right],$$

277 where S, S' are independent copies. Moreover, with $\text{MMD}_{k_{\mathcal{Z}}}(P_{Z|S=s}, P_Z) = \|\mu_{Z|S=s} -$
 278 $\mu_Z\|_{\mathcal{F}_{\mathcal{Z}}}$ the maximum mean discrepancy of §2 under $k_{\mathcal{Z}}$, HSIC is bounded by the
 279 squared expected MMD:

$$280 \quad (3.2) \quad \text{HSIC}(Z, S) \leq \kappa_{\mathcal{S}} \cdot (\mathbb{E}_S[\text{MMD}_{k_{\mathcal{Z}}}(P_{Z|S}, P_Z)])^2.$$

281 *Proof.* See Appendix SM1.6 of the supplement. $\square$

282 **Remark 3.7 (Relation to EIPM).** Equation (3.1) expresses HSIC as a covariance-
 283 weighted aggregation of conditional mean-embedding deviations. Inequality (3.2)
 284 gives only the EIPM-to-HSIC direction: small expected MMD implies small HSIC,
 285 but it does not imply that HSIC upper-bounds EIPM away from zero.

286 The next theorem turns the decomposition (3.1) into a population bridge: small
 287 HSIC controls every RKHS-smoothed average of the conditional mean-embedding
 288 deviation $\Delta(s) := \mu_{Z|S=s} - \mu_Z$.

289 THEOREM 3.8 (Smoothed conditional-distribution control by HSIC). *Let k_Z and*
 290 *k_S be bounded measurable kernels with RKHSs $\mathcal{F}_Z$ and $\mathcal{F}_S$. Let*

$$291 \quad \Delta(s) := \mu_{Z|S=s} - \mu_Z \in \mathcal{F}_Z, \quad \mu_{Z|S=s} := \mathbb{E}[k_Z(Z, \cdot) \mid S = s].$$

292 *Then $\Delta \in L^2(P_S; \mathcal{F}_Z)$ and*

$$293 \quad \text{HSIC}(Z, S) = \|\mathbb{E}_S[\Delta(S) \otimes k_S(S, \cdot)]\|_{\mathcal{F}_Z \otimes \mathcal{F}_S}^2.$$

294 *Moreover, for every $g \in \mathcal{F}_S$,*

$$295 \quad (3.3) \quad \|\mathbb{E}_S[g(S)\Delta(S)]\|_{\mathcal{F}_Z}^2 \leq \|g\|_{\mathcal{F}_S}^2 \text{HSIC}(Z, S).$$

296 *Proof.* See Appendix SM1.7 of the supplement. $\square$

297 COROLLARY 3.9 (Localized sensitive-value contrasts). *Under the assumptions of*
 298 *Theorem 3.8, for any $s_0 \in \mathcal{S}$ with $k_S(s_0, s_0) > 0$,*

$$299 \quad \|\mathbb{E}_S[k_S(S, s_0)(\mu_{Z|S} - \mu_Z)]\|_{\mathcal{F}_Z}^2 \leq k_S(s_0, s_0) \text{HSIC}(Z, S).$$

300 *More generally, for any $s_0, s_1 \in \mathcal{S}$,*

$$301 \quad \|\mathbb{E}_S[\{k_S(S, s_0) - k_S(S, s_1)\}(\mu_{Z|S} - \mu_Z)]\|_{\mathcal{F}_Z}^2 \leq \|k_S(s_0, \cdot) - k_S(s_1, \cdot)\|_{\mathcal{F}_S}^2 \text{HSIC}(Z, S).$$

302 *Proof.* See Appendix SM1.8 of the supplement. $\square$

303 Corollary 3.9 is the population analogue of localized conditioning over nearby
 304 sensitive values. Instead of constructing $\hat{P}_{Z|S=s_0}$, HSIC controls the RKHS mean
 305 embedding of the kernel-weighted signed deviation

$$306 \quad \int k_S(s, s_0) \{P_{Z|S=s} - P_Z\} dP_S(s).$$

307 Thus the joint statistic controls local conditional-distribution contrasts after smooth-
 308 ing by the sensitive-attribute kernel. Pointwise control of $P_{Z|S=s_0}$ would require
 309 additional assumptions on the smoothness of $s \mapsto P_{Z|S=s}$ and on the bandwidth of
 310 k_S .

311 COROLLARY 3.10 (Spectral control of the conditional MMD integral). *Assume*
 312 *the conditions of Theorem 3.8. Let $\Delta(s) := \mu_{Z|S=s} - \mu_Z$ and let $T_S : L^2(P_S) \rightarrow L^2(P_S)$*
 313 *be the kernel integral operator*

$$314 \quad (T_S u)(s) := \int k_S(s, t) u(t) dP_S(t).$$

315 *Let $(\lambda_\ell, \psi_\ell)_{\ell \geq 1}$ be an orthonormal eigensystem of T_S with $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$, and let*
 316 *P_m denote the $L^2(P_S)$ projection onto $\text{span}\{\psi_1, \dots, \psi_m\}$, extended componentwise to*
 317 *$L^2(P_S; \mathcal{F}_Z)$. Then, for any m with $\lambda_m > 0$ such that $\|(I - P_m)\Delta\|_{L^2(P_S; \mathcal{F}_Z)}^2 \leq \rho_m^2$,*

$$318 \quad \mathbb{E}_S[\text{MMD}_{k_Z}^2(P_{Z|S}, P_Z)] = \|\Delta\|_{L^2(P_S; \mathcal{F}_Z)}^2 \leq \frac{1}{\lambda_m} \text{HSIC}(Z, S) + \rho_m^2.$$

319 *Proof.* See Appendix SM1.9 of the supplement. $\square$

Corollary 3.10 gives the nonzero-regime population bridge. HSIC does not uniformly dominate the conditional MMD integral without assumptions, because the kernel integral operator on $\mathcal{S}$ is compact and may have small eigenvalues. What HSIC controls are the conditional-distribution deviations lying in the well-conditioned spectral directions of T_S . If the remaining tail ρ_m is small, then small HSIC implies small conditional-integral MMD. This is the population counterpart of the finite-sample spectral factor $\hat{\lambda}_S^{-1}$ in Theorem 3.12.

COROLLARY 3.11 (Spectral control of GDP for RKHS heads). *Under the assumptions of Corollary 3.10, let $f \in \mathcal{F}_{\mathcal{Z}}$ and define*

$$m_f(s) := \mathbb{E}[f(Z) \mid S = s] - \mathbb{E}[f(Z)].$$

Let P_m be the spectral projection from Corollary 3.10. Then

$$\|P_m m_f\|_{L^2(P_S)}^2 \leq \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{\lambda_m} \text{HSIC}(Z, S).$$

If $\|(I - P_m)m_f\|_{L^2(P_S)}^2 \leq r_{m,f}^2$, then

$$\Delta_{\text{GDP}}(f)^2 = (\mathbb{E}_S |m_f(S)|)^2 \leq \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{\lambda_m} \text{HSIC}(Z, S) + r_{m,f}^2.$$

Proof. See Appendix SM1.10 of the supplement. $\square$

This corollary clarifies the nonzero GDP implication. Small HSIC controls the GDP of an RKHS prediction head up to the spectral tail of its conditional mean function m_f . Without such a spectral or smoothness condition, an aggregate dependence measure does not by itself yield uniform pointwise subgroup control over all $s \in \mathcal{S}$.

Theorem 3.8 is the population nonzero-regime bridge: small HSIC controls every RKHS-smoothed conditional-distribution deviation $\mathbb{E}_S[g(S)\Delta(S)]$. Theorem 3.12 below gives the finite-sample counterpart, controlling the empirical pointwise gap vector $(\hat{\delta}_{f,i})_{i=1}^n$ rather than the smoothed object $\mathbb{E}_S[m_f(S)k_S(S, \cdot)]$.

3.4. Empirical spectral control. The preceding population results show that HSIC controls conditional-distribution deviations after smoothing, and controls the conditional MMD integral along well-conditioned spectral directions of the sensitive-attribute kernel. We now give the finite-sample analogue. The empirical sensitive Gram matrix plays the role of the kernel integral operator T_S , and its positive eigenvalues determine how strongly empirical prediction-deviation directions are controlled.

THEOREM 3.12 (Sensitive-kernel-projected empirical prediction control). *Let $\{(Z_i, S_i)\}_{i=1}^n$ be a sample. Let $K, L \in \mathbb{R}^{n \times n}$ be the Gram matrices*

$$K_{ij} = k_{\mathcal{Z}}(Z_i, Z_j), \quad L_{ij} = k_{\mathcal{S}}(S_i, S_j),$$

and let

$$H := I_n - n^{-1}\mathbf{1}\mathbf{1}^\top, \quad \tilde{K} := HKH, \quad \tilde{L} := H LH.$$

Define the biased empirical HSIC statistic

$$(3.4) \quad \widehat{\text{HSIC}}_n(Z, S) := n^{-2} \text{tr}(\tilde{K}\tilde{L}).$$

Let $r := \text{rank}(\tilde{L})$ and let $P_{\tilde{L}}$ be the orthogonal projection onto the column space of $\tilde{L}$. If $r = 0$, then $P_{\tilde{L}} = 0$ and inequality (3.5) holds trivially with zero left-hand side for every f . If $r \geq 1$, define

$$\hat{\lambda}_S := \lambda_{\min}^+(n^{-1}\tilde{L}),$$

the smallest positive eigenvalue of $n^{-1}\tilde{L}$. For $f \in \mathcal{F}_Z$, define

$$\bar{f}_n := n^{-1} \sum_{j=1}^n f(Z_j), \quad \hat{\delta}_{f,i} := f(Z_i) - \bar{f}_n, \quad \boldsymbol{\delta}_f := (\hat{\delta}_{f,1}, \dots, \hat{\delta}_{f,n})^\top.$$

If $r \geq 1$, then

$$(3.5) \quad \frac{1}{n} \sum_{i=1}^n (P_{\tilde{L}} \boldsymbol{\delta}_f)_i^2 \leq \frac{\|f\|_{\mathcal{F}_Z}^2}{\hat{\lambda}_S} \widehat{\text{HSIC}}_n(Z, S).$$

Proof. See Appendix SM1.11 of the supplement. $\square$

The projection isolates the part of the prediction-deviation vector that lies in the directions on which $\tilde{L}$ has positive eigenvalues; the kernel of $\tilde{L}$ contains the constant direction $\mathbf{1}$ but may also contain additional directions arising from repeated or ordinal S values.

COROLLARY 3.13 (Unprojected bound under full rank). *Adopt the notation and assumptions of Theorem 3.12. If $\tilde{L}$ has rank $n-1$, then the bound (3.5) holds without the projection,*

$$\frac{1}{n} \sum_{i=1}^n \hat{\delta}_{f,i}^2 \leq \frac{\|f\|_{\mathcal{F}_Z}^2}{\hat{\lambda}_S} \widehat{\text{HSIC}}_n(Z, S),$$

and, for the empirical absolute prediction gap $\hat{\Delta}_f := n^{-1} \sum_{i=1}^n |\hat{\delta}_{f,i}|$,

$$\hat{\Delta}_f^2 \leq \frac{1}{n} \sum_{i=1}^n \hat{\delta}_{f,i}^2 \leq \frac{\|f\|_{\mathcal{F}_Z}^2}{\hat{\lambda}_S} \widehat{\text{HSIC}}_n(Z, S).$$

Proof. When $\text{rank}(\tilde{L}) = n-1$, the column space of $\tilde{L}$ is the orthogonal complement of $\text{span}\{\mathbf{1}\}$. Since $\sum_{i=1}^n \hat{\delta}_{f,i} = 0$, the vector $\boldsymbol{\delta}_f$ lies in that complement, so $P_{\tilde{L}} \boldsymbol{\delta}_f = \boldsymbol{\delta}_f$ and (3.5) holds without the projection. The absolute-gap bound follows by the Cauchy-Schwarz inequality. $\square$

Interpretation. Theorem 3.12 and Corollary 3.13 give a finite-sample spectral bridge from empirical HSIC to empirical prediction deviations. The centered sensitive Gram matrix $\tilde{L}$ determines which directions of variation over the observed sensitive values are controlled: HSIC controls the projection of the centered prediction-deviation vector onto the column space of $\tilde{L}$, and when $\text{rank}(\tilde{L}) = n-1$, as for a strictly positive definite kernel such as the Gaussian kernel on distinct sample points, this projection is the identity on centered vectors and the bound becomes unprojected. The quantities $\hat{\delta}_{f,i}$ are centered empirical prediction deviations, not smoothed estimators of $\mathbb{E}[f(Z) \mid S = s]$.

4. FRHSIC: Empirical Objective and Statistical Control.

4.1. Empirical FRHSIC objective. FRHSIC trains the representation $Z := h(X)$ (as in §2) by penalizing empirical dependence between Z and S over a feasible encoder class $\mathcal{H}$ and prediction-head class $\mathcal{F}$. With fixed bounded characteristic kernels k_Z, k_S , the empirical HSIC penalty $\widehat{\text{HSIC}}_n(h(X), S)$ is the biased V -statistic (3.4) formed from the paired sample $\{(h(X_i), S_i)\}_{i=1}^n$. FRHSIC solves the empirical regularized problem

$$(4.1) \quad \min_{h \in \mathcal{H}, f \in \mathcal{F}} \hat{\mathcal{L}}_n(f \circ h) + \lambda \widehat{\text{HSIC}}_n(h(X), S),$$

where $\widehat{\mathcal{L}}_n$ is the empirical prediction loss (cross-entropy for classification, MSE for regression) and $\lambda > 0$ controls the fairness–accuracy tradeoff. Equation (4.1) is the Lagrangian relaxation of the empirical constrained problem $\min_{h,f} \widehat{\mathcal{L}}_n(f \circ h)$ subject to $\widehat{\text{HSIC}}_n(h(X), S) \leq \delta$. We use the biased estimator because it is nonnegative and hence a valid penalty; the unbiased U -statistic estimator (Song et al., 2012) can be negative in finite samples, though both are consistent.

The penalty is a joint-sample statistic on $\{(h(X_i), S_i)\}$ and does not construct local estimates of $P_{Z|S=s}$. The concentration result below is stated for fixed kernels. In the experiments the kernel bandwidths are set by the median heuristic and the resulting statistic is used as a practical training penalty; the theorem isolates the fixed-kernel statistical behavior.

4.2. Uniform train-to-population control. The empirical penalty is evaluated on an encoder selected from the same sample, so a pointwise concentration bound for a fixed encoder is not sufficient; the relevant statement must hold uniformly over the encoder class $\mathcal{H}$.

THEOREM 4.1 (Uniform concentration of empirical HSIC over the encoder class).

Let $\{(X_i, S_i)\}_{i=1}^n$ be i.i.d. copies of (X, S) , and let $\mathcal{H}$ be a class of encoders $h : \mathcal{X} \rightarrow \mathcal{Z} \subseteq \mathbb{R}^{d_Z}$. For $h \in \mathcal{H}$ write $Z^h := h(X)$ and $Z_i^h := h(X_i)$. Let k_Z, k_S be fixed kernels with feature maps $\phi_Z(z) := k_Z(z, \cdot) \in \mathcal{F}_Z$ and $\phi_S(s) := k_S(s, \cdot) \in \mathcal{F}_S$, satisfying the boundedness and Lipschitz conditions

$$\sup_{z \in \mathcal{Z}} k_Z(z, z) \leq \nu_Z, \quad \sup_{s \in \mathcal{S}} k_S(s, s) \leq \nu_S,$$

$$\|\phi_Z(z) - \phi_Z(z')\|_{\mathcal{F}_Z} \leq \ell_Z \|z - z'\|, \quad \|\phi_S(s) - \phi_S(s')\|_{\mathcal{F}_S} \leq \ell_S \|s - s'\|,$$

where $k_Z(z, z) = \|\phi_Z(z)\|_{\mathcal{F}_Z}^2$ and $k_S(s, s) = \|\phi_S(s)\|_{\mathcal{F}_S}^2$. For $h \in \mathcal{H}$, write $\widehat{\text{HSIC}}_n(h)$ for the biased empirical HSIC estimator (3.4) computed from the paired sample $\{(Z_i^h, S_i)\}_{i=1}^n$, and $\text{HSIC}(h)$ for the population HSIC of Definition 2.1 with $Z = Z^h$; these are the estimator and functional already defined, evaluated at the encoder-transformed representation Z^h with the fixed kernels k_Z, k_S unchanged. Let

$$\widehat{\mathcal{G}}_n(\mathcal{H}) := \mathbb{E}_\xi \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \langle \xi_i, h(X_i) \rangle \mid X_1, \dots, X_n \right], \quad \xi_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_{d_Z}),$$

be the empirical Gaussian complexity of $\mathcal{H}$, and set $\mathcal{G}_n(\mathcal{H}) := \mathbb{E}[\widehat{\mathcal{G}}_n(\mathcal{H})]$. Define

$$(4.2) \quad B_n(\mathcal{H}, \delta) := 8\sqrt{2} \nu_Z \nu_S \sqrt{\frac{\log(2/\delta)}{n}} + \frac{4\nu_Z \nu_S}{n} + 48\sqrt{\pi} \max\{\nu_Z \ell_S, \nu_S \ell_Z\} \mathcal{G}_n(\mathcal{H}).$$

Then, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$(4.3) \quad \sup_{h \in \mathcal{H}} |\widehat{\text{HSIC}}_n(h) - \text{HSIC}(h)| \leq B_n(\mathcal{H}, \delta).$$

Proof. See Appendix SM7.2 of the supplement. The proof maps FRHSIC to the uniform HSIC concentration theorem of Ni and Huo (2024): for each h , the empirical HSIC (3.4) at $\{(Z_i^h, S_i)\}_{i=1}^n$ is the biased estimator for the product kernel $\kappa_h((x, s), (x', s')) := k_Z(h(x), h(x')) k_S(s, s')$; the sensitive coordinate is untransformed, so the singleton class on $\mathcal{S}$ contributes zero Gaussian complexity; and the cited result applied to $\{\kappa_h : h \in \mathcal{H}\}$ yields (4.3) with the constants of (4.2) (Proposition SM7.2). $\square$

COROLLARY 4.2 (Population HSIC control for the learned encoder). *Suppose the assumptions of Theorem 4.1 hold, and let $\hat{h} \in \mathcal{H}$ be any data-dependent encoder selected from the same sample, including one obtained by minimizing the FRHSIC objective (4.1). Then, with probability at least $1 - \delta$,*

$$\text{HSIC}(\hat{h}) \leq \widehat{\text{HSIC}}_n(\hat{h}) + B_n(\mathcal{H}, \delta).$$

Proof. The event in Theorem 4.1 holds simultaneously for every $h \in \mathcal{H}$. Since $\hat{h} \in \mathcal{H}$, evaluating (4.3) at $h = \hat{h}$ gives the claim. $\square$

Corollary 4.2 is the train-to-population bridge: small empirical HSIC for the selected encoder implies small population dependence of the learned representation, up to the uniform concentration error, independently of the stochastic optimization used to minimize (4.1).

COROLLARY 4.3 (Root- n rate under controlled encoder complexity). *Under the assumptions of Theorem 4.1, suppose $\mathcal{G}_n(\mathcal{H}) = O(n^{-1/2})$. Then, for every fixed $\delta \in (0, 1)$,*

$$\sup_{h \in \mathcal{H}} |\widehat{\text{HSIC}}_n(h) - \text{HSIC}(h)| = \tilde{O}_p(n^{-1/2}),$$

where $\tilde{O}$ hides logarithmic factors, and the same rate controls the learned encoder $\hat{h}$ through Corollary 4.2.

Proof. Substituting $\mathcal{G}_n(\mathcal{H}) = O(n^{-1/2})$ into (4.2) gives $B_n(\mathcal{H}, \delta) = \tilde{O}(n^{-1/2})$; the claim then follows from (4.3) and Corollary 4.2. $\square$

Encoder classes covered by the rate. The condition $\mathcal{G}_n(\mathcal{H}) = O(n^{-1/2})$ is a fixed-effective-complexity condition on the feasible encoder class. For bounded linear encoders $\mathcal{H}_{\text{lin}} = \{h_W(x) = Wx : \|W\|_F \leq R\}$ with $\|X\| \leq M$ almost surely,

$$\widehat{\mathcal{G}}_n(\mathcal{H}_{\text{lin}}) \leq R \left\| \frac{1}{n} \sum_{i=1}^n \xi_i X_i^\top \right\|_F, \quad \text{hence} \quad \mathcal{G}_n(\mathcal{H}_{\text{lin}}) \leq RM \sqrt{d_Z/n},$$

which is $O(n^{-1/2})$ for fixed representation dimension d_Z . The same $n^{-1/2}$ order holds for fixed-depth, fixed-width MLP encoders with bounded weights, bounded inputs, and Lipschitz activations, and for finite-dimensional parametric families over compact parameter sets, under the corresponding complexity bounds of Ni and Huo (2024). The rate does not extend to arbitrary unregularized or growing-width networks.

Implementation. Training optimizes (4.1) with minibatch stochastic gradient methods: on a minibatch of size m the HSIC penalty is the same trace formula (3.4) restricted to the batch, at $O(m^2)$ cost. Kernel bandwidths use the median heuristic. The penalty extends to Equal Opportunity by restricting the statistic to the $Y = 1$ subset and to multiple sensitive attributes through a product kernel on the sensitive space, and λ is chosen by held-out validation; details are in Appendix SM5.2 and Appendix SM8.6 of the supplement.

5. Experiments. We evaluate FRHSIC along four questions. §5.2 checks whether the empirical estimator follows the predicted convergence rate. §5.3 compares its fairness–accuracy tradeoffs against the baselines on real data. §5.4 tests whether the enforced fairness transfers to fresh downstream prediction heads. §5.5 measures training time relative to conditional-route methods.

Code, cached results, and figures are available in a public repository at <https://github.com/Yijin911/FRHSIC>, with an archived DOI release accompanying the camera-ready version.

5.1. Experimental setup. We use five real datasets with continuous sensitive attributes. Adult, ACS Income, MEPS, and COMPAS are classification tasks scored by accuracy; Crime is a regression task scored by mean squared error. Fairness is measured by held-out Δ_{GDP} , with lower values indicating less dependence on the sensitive attribute S .

- **Adult:** $n \approx 30,000$ individuals; income $>50\text{K}$; sensitive attribute age.
- **ACS Income:** $n = 20,000$ from the 2018 California American Community Survey via folktables (Ding et al., 2021); income $>50\text{K}$; age.
- **MEPS:** $n \approx 13,000$ from the 2015 Medical Expenditure Panel Survey (Romano et al., 2020); healthcare utilization ≥ 10 visits; age.
- **Crime** (the Communities and Crime dataset): $n \approx 2,000$ communities; violent crimes per population; racial composition.
- **COMPAS:** $n \approx 6,000$ defendants; two-year recidivism; age.

Features and S are min–max scaled to $[0, 1]$, fit on the training split only. We compare FRHSIC with the continuous-sensitive methods FREM (Kong et al., 2025) and Reg-GDP (Jiang et al., 2022), the adversarial method ADV (Grari et al., 2022), the binary-attributhe methods MMD (Deka and Sutherland, 2023) and LAFTR (Madras et al., 2018) applied to quartile-binned S , and an unconstrained baseline (Unfair, $\lambda = 0$). Following Kong et al. (2025), all methods share a two-layer SELU encoder with hidden and representation dimension 50 and a linear prediction head, train for 200 epochs with Adam at learning rate 10^{-3} , and are swept over $\lambda \in \{0.1, 1, 10, 100, 500\}$ with results averaged over five random 80/20 splits, reported as mean $\pm$ standard deviation. Each method keeps the kernel bandwidth convention of its source publication; we do not retune bandwidths to favor any method, and the only fairness-specific hyperparameter FRHSIC adds is λ , selected by the held-out procedure of Section SM8.6. Full preprocessing, hyperparameter grids, hardware, and the FREM 32-anchor subsampling approximation are documented in Section SM4 of the supplement.

5.2. Synthetic estimator behavior. The empirical estimator follows the rate the theory predicts. We draw (Z, S) jointly Gaussian with correlation $\rho = 0.5$ ($Z \sim \mathcal{N}(0, 1)$, $S = \rho Z + \sqrt{1 - \rho^2} \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 1)$), take the $n = 20,000$ statistic as ground truth, and report the absolute error of each estimator at sample sizes from 50 to 5,000, averaged over 20 replications with Gaussian kernels at bandwidth 1. The EIPM estimator is the weighted MMD estimator of Kong et al. (2025), the same estimator used by the FREM baseline. On this dependence, $\overline{\text{HSIC}}$ converges at the $O(n^{-1/2})$ rate that underlies the uniform train-to-population control of §4.2, while the EIPM estimator converges at the slower $O(n^{-2/5})$ rate. Figure 2 reports both estimation errors against sample size; the two estimators target different population functionals that coincide only on their shared zero set, so the comparison concerns estimator behavior rather than equality of objectives. The conditional-gap bound of Theorem 3.12 is separately validated on a synthetic classification setup in Appendix SM5.1 of the supplement.

5.3. Fairness–accuracy tradeoffs. FRHSIC achieves fairness–accuracy tradeoffs comparable to the strongest continuous-sensitive baselines, although it is not uniformly best on any dataset. Because HSIC and GDP penalties have different scales, equal λ does not mean equal fairness, so Table 1 compares at a matched operating point: for each method we report the lowest Δ_{GDP} among models whose prediction performance stays within 1% of the Unfair baseline, together with the bandwidth-free normalized area under the accuracy–fairness Pareto envelope, nAUP. Figures 3 and 4

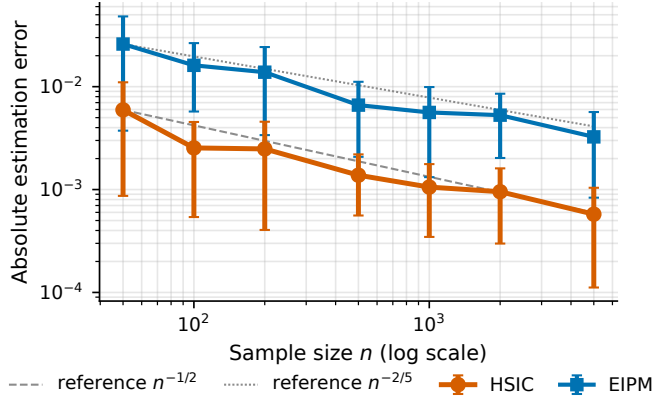


FIG. 2. *Synthetic estimator convergence. Absolute estimation error is plotted against sample size on log-log axes. Empirical HSIC follows the predicted $O(n^{-1/2})$ behavior and the smoothing-based EIPM estimator follows the slower $O(n^{-2/5})$ trend: the fitted log-log slopes are -0.46 for HSIC and -0.44 for EIPM, close to the predicted $-1/2$ and $-2/5$. The dashed and dotted gray lines are the $n^{-1/2}$ and $n^{-2/5}$ references. The two estimators target different population functionals that coincide only on their zero set.*

show the full frontiers over the λ sweep, for the classification datasets and for Crime respectively.

No method dominates across datasets. By nAUP, FRHSIC is best on COMPAS, within .004 of the best on Crime, and mid-pack on Adult, ACS Income, and MEPS, staying within the spread of FREM, Reg-GDP, and the adversarial baselines. The binned baselines MMD and LAFTR are generally worse than the continuous-sensitive methods, which supports treating S as continuous rather than discretizing it. The matched band also removes a Reg-GDP artifact: Reg-GDP reaches near-zero Δ_{GDP} only by collapsing accuracy outside the 1% band, from .849 to .758 on Adult and from .790 to .590 on ACS Income at its unconstrained point. Differences within ± 0.01 accuracy or ± 0.05 on Δ_{GDP} fall inside the five-split spread and should be read as practically equivalent. COMPAS age is integer-valued with few distinct values, so its centered sensitive Gram matrix is low rank, the regime described by Theorem 3.12.

5.4. Transfer to fresh downstream heads. Because HSIC penalizes all dependence between Z and S , any head built on the representation should inherit the fairness property, whereas a method that regularizes a single training head need not transfer. We freeze the learned representation of each method at a fixed λ and train four fresh heads on it: linear, two-layer MLP, random forest, and SVM. Table 2 reports the standard deviation of Δ_{GDP} across the four heads, a stability diagnostic in which lower values indicate fairness less tied to the head used during training. FRHSIC’s cross-head variability is comparable to FREM and well below the Unfair baseline. Reg-GDP attains lower variability on Adult, but only at the accuracy collapse noted in §5.3. This is a stability diagnostic rather than a guarantee that every downstream head is fair; the full per-head breakdown is in Section SM8.2 of the supplement.

5.5. Computational efficiency. FRHSIC trains substantially faster than FREM because it replaces FREM’s per-sample weighted conditional-MMD computation with a single $O(m^2)$ HSIC trace per minibatch. Figure 5 reports per-epoch wall-clock time on synthetic data of varying size, at batch size 256 from $n = 500$

TABLE 1

Matched-performance comparison on the five real datasets. For each method we report the lowest Δ_{GDP} among models whose performance stays within 1% of the Unfair baseline. Perf is accuracy, except Crime where lower $\text{MSE} \times 10^2$ is better; Δ_{GDP} lower is fairer; nAUP is the normalized area under the accuracy-fairness Pareto envelope, higher is better. Bold marks the best fairness method per column; “—” means no swept λ meets the band. Setup and bandwidth conventions are in §5.1.

Dataset	Method	Perf.	Δ_{GDP} ($\downarrow$)	nAUP ($\uparrow$)
Adult (Acc $\uparrow$)	Unfair	.849	.749	—
	FRHSIC (Ours)	.845	.232	.786
	FREM	.845	.167	.835
	Reg-GDP	.842	.082	.872
	ADV	.841	.466	.434
	MMD (binned)	.847	.126	.880
	LAFTR (binned)	.842	.385	.555
ACS Income (Acc $\uparrow$)	Unfair	.790	.514	—
	FRHSIC (Ours)	.783	.228	.805
	FREM	.787	.274	.865
	Reg-GDP	.787	.292	.852
	ADV	.785	.458	.713
	MMD (binned)	.783	.157	.881
	LAFTR (binned)	.787	.474	.754
MEPS (Acc $\uparrow$)	Unfair	.804	.626	—
	FRHSIC (Ours)	.797	.329	.464
	FREM	.800	.317	.549
	Reg-GDP	.801	.370	.469
	ADV	.803	.480	.487
	MMD (binned)	.800	.407	.492
	LAFTR (binned)	.803	.530	.477
Crime (MSE $\downarrow$)	Unfair	1.91	.029	—
	FRHSIC (Ours)	1.91	.029	.686
	FREM	1.87	.023	.680
	Reg-GDP	1.91	.025	.690
	ADV	1.92	.029	.002
	MMD (binned)	—	—	.573
	LAFTR (binned)	1.89	.029	.103
COMPAS (Acc $\uparrow$)	Unfair	.654	.123	—
	FRHSIC (Ours)	.649	.090	.679
	FREM	.648	.101	.530
	Reg-GDP	.650	.079	.480
	ADV	.649	.095	.314
	MMD (binned)	.651	.082	.485
	LAFTR (binned)	.655	.122	.475

to $n = 20,000$, with all methods sharing the encoder and prediction head. At $n = 20,000$ and batch size 256, FRHSIC completes an epoch in 2.82s versus 100.7s for FREM, a roughly $36\times$ speedup. Reg-GDP is faster still at 1.69s because it evaluates conditional expectations on a fixed grid rather than kernel matrices, but it optimizes a head-specific smoothed moment criterion rather than a representation-level distributional penalty; the relevant comparison for FRHSIC is therefore FREM, the most directly comparable conditional-route method. The full-batch $O(n^2)$ versus $O(n^3)$

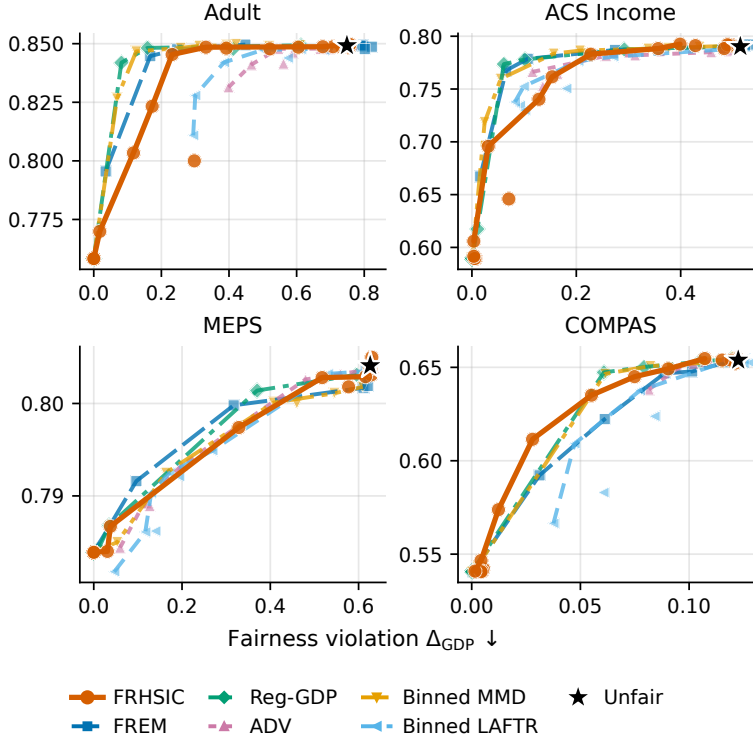


FIG. 3. Fairness-accuracy frontiers on the classification datasets. Each point corresponds to one value of the fairness regularization parameter λ ; lower Δ_{GDP} is fairer and higher accuracy is better. Lines connect the Pareto-efficient points of each method. Axes are scaled separately by dataset to show the frontier structure. The black star marks the Unfair baseline, the unconstrained model trained without any fairness penalty ($\lambda = 0$). FRHSIC is competitive with continuous-sensitive baselines across datasets, but no method uniformly dominates. Table 1 gives the matched-performance summary.

TABLE 2

Cross-head Δ_{GDP} standard deviation across four fresh heads (linear, MLP, random forest, SVM) trained on a single frozen representation at $\lambda = 10$ (Adult) and $\lambda = 100$ (Crime); lower indicates fairness less tied to the training head. Full per-head results are in Section SM8.2 of the supplement.

Method	Adult ($\lambda = 10$)	Crime ($\lambda = 100$)
Unfair	0.348	—
FRHSIC (Ours)	0.244	0.001
FREM	0.247	0.002
Reg-GDP	0.058 [†]	0.002
ADV	0.252	0.005

[†] Reg-GDP attains lower cross-head variance on Adult but with accuracy collapsing to 0.758 (Table 1).

analysis is given in Section SM4 of the supplement.

Additional experiments are reported in Section SM8.2–SM8.6 of the supplement: an empirical tightness sweep for the Theorem 3.12 bound, a high-dimensional

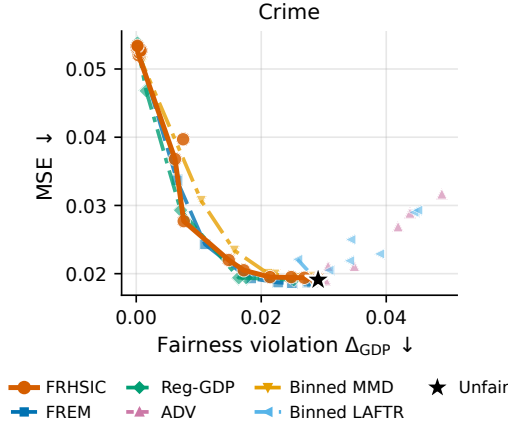


FIG. 4. *Fairness-prediction-error frontier on Crime. Each point corresponds to one value of λ ; lower Δ_{GDP} is fairer and lower MSE is better. Lines connect the Pareto-efficient points of each method; the black star marks the Unfair baseline ($\lambda = 0$, no fairness penalty). The separate panel avoids mixing regression error with classification accuracy.*

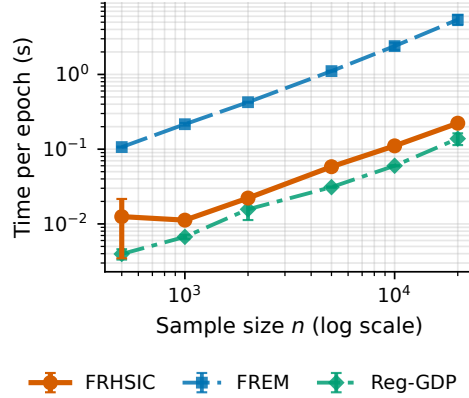


FIG. 5. *Per-epoch wall-clock time versus sample size at batch size 256, on log-log axes; error bars are ± 1 standard deviation over repeats and are negligible except for FRHSIC at the smallest n . FRHSIC uses one HSIC trace computation per batch and trains substantially faster than FREM’s weighted-MMD computation: at $n = 20,000$, FRHSIC takes 2.82s per epoch versus 100.7s for FREM, a roughly $36\times$ speedup. Reg-GDP is faster but optimizes a head-specific smoothed criterion rather than a representation-level distributional penalty.*

sensitive-attribute comparison, a mutual-information diagnostic, ablations over kernel choice and representation dimensionality, and validation of the λ -selection rule of Section SM8.6.

6. Related Work.

Fair representation learning. Zemel et al. (2013) introduced the framework of mapping data to a latent space that preserves task information while obfuscating S . Subsequent work extended this through variational autoencoders (Louizos et al., 2015), adversarial training (Madras et al., 2018), disentangled representations (Creager et al., 2019), MMD- and Sinkhorn-based formulations (Cho et al., 2020a; Deka

and Sutherland, 2023), and data-domain transformations (Quadrianto et al., 2019). Most of these methods are designed for categorical sensitive attributes and discretize continuous S at training time.

Continuous-sensitive-attribute fairness criteria. GDP (Jiang et al., 2022) is estimated via Nadaraya–Watson kernel smoothing on S . EIPM/FREM (Kong et al., 2025) extends GDP to higher moments via a weighted empirical MMD with bandwidth γ_n and leave-one-out correction. Kernel HGR / NLD (Mary et al., 2019; Grari et al., 2022) estimates a global dependence criterion via regularized kernel canonical correlation. Mutual information (Cho et al., 2020b) is estimated via density or copula approximations. Theorem 3.3 of §3.2 disintegrates the joint-vs-product IPM into a P_S -weighted conditional contrast, and Corollary 3.5 recovers the conditional-integral IPM that GDP and EIPM instantiate when the witness class decomposes across s ; kHGR is a supremum-of-correlations criterion outside this form.

HSIC and dependence measures. HSIC was introduced by Gretton et al. (2005) and has been used for independence testing (Gretton et al., 2007; Albert et al., 2022), feature selection (Song et al., 2007, 2012), dimensionality reduction (Ma et al., 2018), and as a training objective for deep networks (Ma et al., 2020). The equivalence of HSIC with distance covariance (Sejdinovic et al., 2013) situates the present work within a single family of joint-distribution dependence measures.

7. Discussion.

The joint-distribution view. GDP, EIPM, and mutual information are conditional-integral criteria, while HSIC is a joint-vs-product discrepancy. Theorem 3.3 connects the two routes by disintegrating the joint discrepancy into a P_S -weighted conditional contrast, and Corollary 3.5 recovers the conditional-integral IPM under decomposable witness classes. The joint route thus preserves the same independence target while changing the object that is estimated: the HSIC instance has a closed-form V -statistic and controls smoothed conditional deviations and, under spectral regularity, the conditional MMD integral up to a spectral tail (Theorem 3.12, Corollary 3.13).

Extensions. The framework extends to representation-level Equal Opportunity by computing the joint discrepancy on the $Y = 1$ subset, and to multiple continuous sensitive attributes by taking a product kernel on $\mathcal{S}_1 \times \mathcal{S}_2$. Both extensions appear in Appendix SM4. Replacing the reference distribution $P_Z \otimes P_S$ by $P_Z \otimes \text{Unif}(\mathcal{S})$ yields a minority-protection target in which rare values of S receive equal weight; this is a modeling choice and does not affect the structural identity of Theorem 3.3.

Open questions. Two questions remain. First, the empirical bound in Theorem 3.12 scales as $\|f\|_{\mathcal{F}_Z}^2 / \hat{\lambda}_S$; tighter spectral arguments exploiting the full singular-value distribution of $\tilde{L}$ would yield sharper guarantees on the integral form. Second, the framework treats the joint discrepancy with an RKHS witness class; non-RKHS instances (Wasserstein, total variation) inherit the structural identity but lose the parametric rate and closed-form estimator. We leave these to future work.

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